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**SECI1143 PROBABILITY & STATISTICAL DATA ANALYSIS
SECTION 01**

**PROJECT 2
HEART DISEASE DATA ANALYSIS**

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1.0 Introduction

Heart disease is also known as cardiovascular disease. It encompasses a range of conditions that affect the heart and blood vessels. Nowadays, heart disease has become a common cause of death. There's 1 in every 5 deaths caused by heart disease. It is one of the leading causes of death worldwide, making its analysis critical for understanding, preventing, and treating the disease. Heart disease refers to the diseases of the heart, for example, coronary artery disease, heart failure, heart valve abnormalities, and abnormal heart rhythms. Heart disease is a significant public health challenge with complex causes and serious implications. Through understanding risk factors, symptoms, diagnostic methods, and treatment options, individuals and healthcare providers can work together to prevent and manage heart disease. Public health initiatives focusing on lifestyle changes, early detection, and effective treatment are essential in reducing the impact of cardiovascular disease globally. The objective of this project is to identify the risk factor, to determine which variables are significantly associated with heart disease. Next is also to predict the outcomes by developing predictive models to estimate the likelihood of heart disease based on the individual risk profiles. From the data, we expect to see that the heart disease is significantly affected by age, sex, maximum heart rate, resting blood pressure, and type of chest pain.

2.0 Dataset

The dataset that we have chosen for our project is Heart Disease Data Analysis. This dataset contains information on patients with their heart conditions. The variables in the dataset are shown in Table 1 and a summary of selected variables and their respective data types is demonstrated in Table 2.

Name	Variable	Data Type
Age	Age of the patient	Ratio
Sex	Sex of the patient (1 = male, 0 = female)	Nominal
Chest Pain Type	Chest pain type (0-3)	Ordinal
Resting Blood Pressure	Resting blood pressure (in mm Hg)	Ratio
Maximum Heart Rate	Maximum heart rate achieved	Ratio

Table 1: Variables in heart disease data

Selected Variable(s)	Test	Description
Age, Sex	Hypothesis testing (two sample test)	Explanation: The variable is used to test whether there is a significant difference of the average age between males and females at 5% significance level. Possible outcome: There is a significant difference between the average age of males and females at 5% significance level.
Age, Maximum heart rate	Correlation test	Explanation:The variables are selected to test whether the linear relationship exists between the age and the maximum heart rate using Pearson's Product-Moment Correlation Coefficient at 5% of significance level. Possible outcome: There is a negative linear relationship between the age and the maximum heart rate at 5% of significance level. The higher the age, the lower the maximum heart rate.
Resting blood pressure, Maximum heart rate	Regression Test	Explanation: The variables are selected to test whether the value of resting blood pressure depends on the maximum heart rate, using resting blood pressure as dependent variable x and maximum heart rate as independent variable y at 5% significance level. Possible outcome: The resting blood pressure depends on maximum heart rate at 5% significance level. The higher the maximum heart rate, the higher the resting blood pressure.
Sex, Chest Pain Type	Chi-Square Test of Independence	Explanation: The variables are selected to test whether there is association between sex and chest pain type at 5% significance level. Possible outcome: Sex of patient is not independent of chest pain type experienced by patient at 5% significance level.

Table 2: Summary of selected variables and test with respective description

3.0 Data Analysis

3.1 Hypothesis Testing - 2 samples

By using hypothesis tests, we wish to compare the average age between males and females to determine if there is a significant difference. This will be achieved by conducting a two-sample t-test using the significance level(α) of 0.05.

Hypothesis statement:

Null hypothesis (H_0) : $\mu_{\text{male}} = \mu_{\text{female}}$ (There is no difference in average age between males and females.)

Alternative hypothesis (H_1) : $\mu_{\text{male}} \neq \mu_{\text{female}}$ (There is a difference in average age between males and females)

Significance level(α) : 0.05

Test statistic:

$$T_0^* = \frac{\bar{X}_1 - \bar{X}_2 - \Delta_0}{\sqrt{\frac{S_1^2}{n1} + \frac{S_2^2}{n2}}}$$

Degree of Freedom:

$$v = \frac{\left(\frac{S_1^2}{n1} + \frac{S_2^2}{n2} \right)^2}{\frac{\left(\frac{S_1^2}{n1} \right)^2}{n1-1} + \frac{\left(\frac{S_2^2}{n2} \right)^2}{n2-1}}$$

By using R studio, we get the result :

```
> # Print the results
> print(t_test_result)

Welch Two Sample t-test

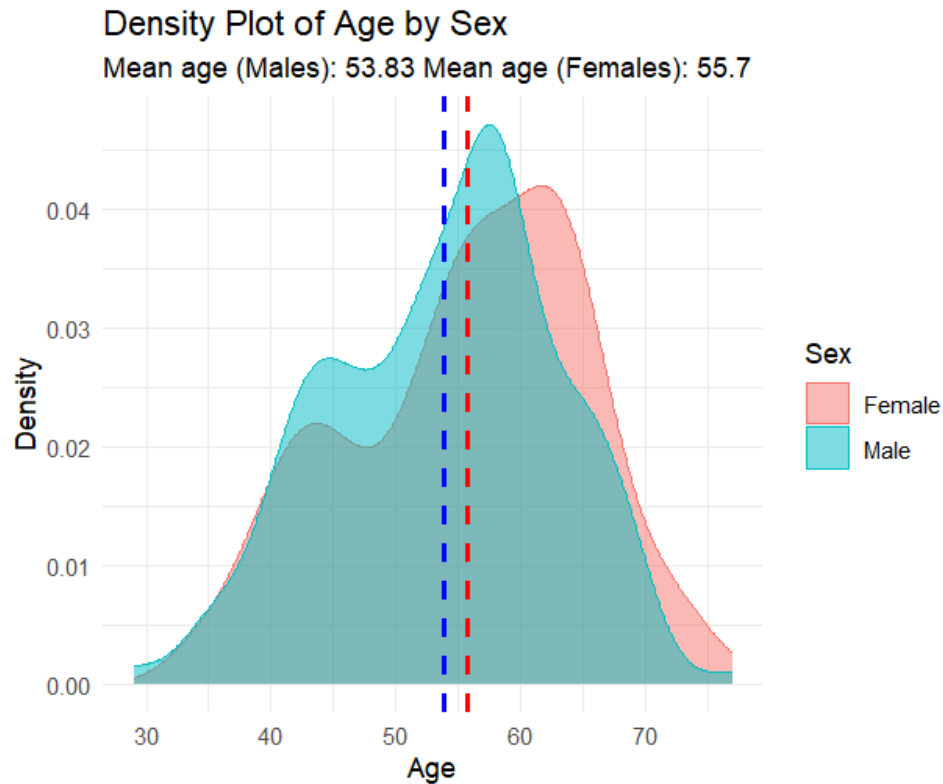
data: male_ages and female_ages
t = -2.6213, df = 472.76, p-value = 0.009041
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -3.260450 -0.466584
sample estimates:
mean of x mean of y
 53.83333  55.69685

>
> # Print t-value and critical value
> cat("T-value:", t_value, "\n")
T-value: -2.621315
> cat("Critical value (two-tailed test, alpha = 0.05):", critical_value, "\n")
Critical value (two-tailed test, alpha = 0.05): 1.964995
>
> # Compare t-value with critical value
> if (abs(t_value) > critical_value) {
+   cat("Decision: Reject the null hypothesis\n")
+ } else {
+   cat("Decision: Fail to reject the null hypothesis\n")
+ }
Decision: Reject the null hypothesis
```

From the result, we know that the t-value is -2.621315. And since $\alpha = 0.05$ and it's a two tailed test, we get the value of $t_{0.025, 473} = 1.964995$ by using R studio. Hence, we will reject the null hypothesis, H_0 if $t < -1.964995$ or $t > 1.964995$.

Decision:

Since $t = -2.621315 < -1.964995$, we reject H_0 . There is sufficient evidence to conclude that there is a significant difference between male and female ages. at the significance level, $\alpha = 0.05$.



Through using R studio, we also created a graph to compare the average age between males and females to determine if there is a significant difference. Since this is a two sample hypothesis test, the density plot presents age distributions for males which represent in blue colour and females which represent in red colour in the graph above. This graph allows for a visual assessment of differences in age distribution shapes and central tendencies. Moreover, the blue dot-line is representing the mean value of male age (53.83) whereas the red dot-line is representing the mean value of female age (55.70) . From this graph, it clearly shows that there is a significant difference between male and female ages.

3.2 Correlation Test

We had used the variables of age and maximum heart rate achieved in this correlation test. In this test, we are going to test whether there is a linear relationship between the age and the maximum heart rate achieved using the significance level(α) of 0.05.

Pearson's Product-Moment Correlation Coefficient:

$$r = \frac{\Sigma xy - \frac{(\Sigma x \Sigma y)}{n}}{\sqrt{[(\Sigma x^2) - \frac{(\Sigma x)^2}{n}][(\Sigma y^2) - \frac{(\Sigma y)^2}{n}]}}$$

Hypothesis statement:

$H_0 : \rho = 0$ (No linear correlation)

$H_1 : \rho \neq 0$ (Linear correlation exists)

Test statistic:

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$$

```
> print(cor_test_result)
```

```
Pearson's product-moment correlation
```

```
data: x and y
t = -12.412, df = 798, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.4587918 -0.3425061
sample estimates:
      cor
-0.4022702
```

By using R studio, we get the value of the test statistic, $t = -12.412$.

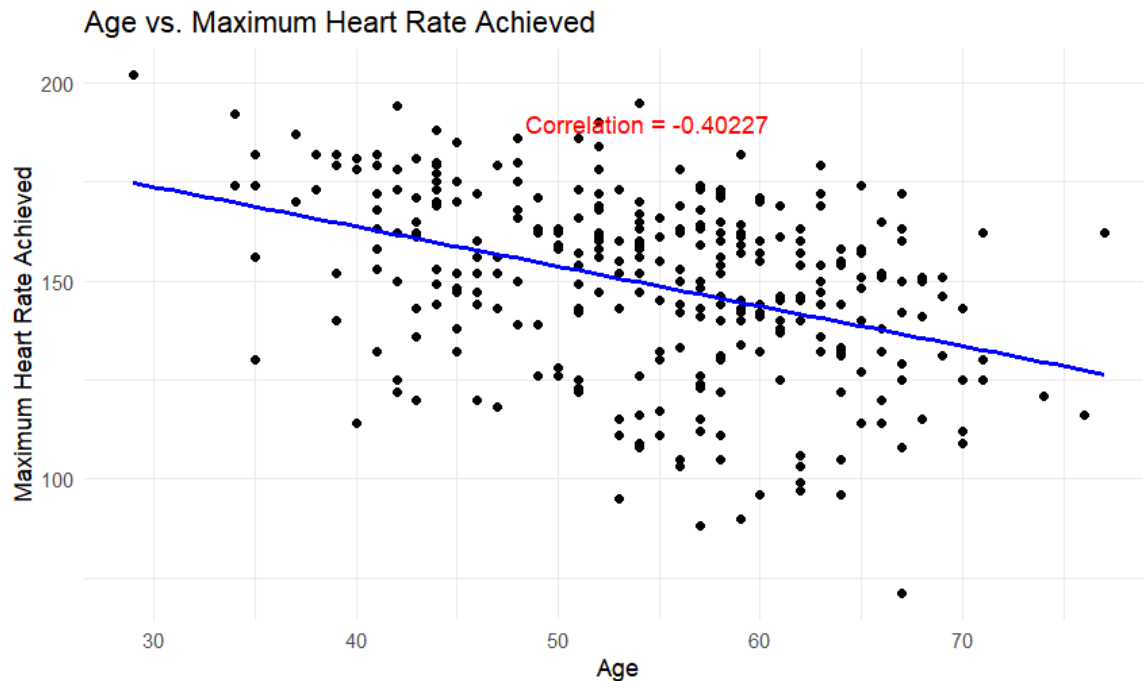
The value of degree of freedom, $\nu = n - 2 = 800 - 2 = 798$.

```
> print(paste("The critical value for df =", df, "and alpha =", alpha * 2, "is:", t_value))
[1] "The critical value for df = 798 and alpha = 0.05 is: 1.96294119282661"
```

Since $\alpha = 0.05$ and it's a two tailed test, we get the value of $t_{0.025, 798} = 1.96294$ by using R studio. Hence, we will reject the null hypothesis, H_0 if $t < -1.96294$ or $t > 1.96294$.

Decision:

Since $t = -12.412 < -1.96294$, we reject H_0 . There is sufficient evidence to conclude that there is a linear relationship between the age and the maximum heart rate achieved at the significance level, $\alpha = 0.05$.



Graph 2: maximum heart rate achieved against age

Based on the scatter plot graph above, we can observe a slight downward slope. It has been denoted that there is a negative relationship between the age and the maximum heart rate achieved. This suggests that as people age, their maximum heart rate tends to decrease.

```
> # Print the correlation coefficient
> cat("Pearson correlation coefficient is:", correlation, "\n")
Pearson correlation coefficient is: -0.4022702
```

We used Pearson's product-moment correlation coefficient to calculate the correlation. Using R studio, we found that the value of $r = -0.40227$, which indicates a weak negative linear relationship between the age and the maximum heart rate achieved.

The scatter plot graph and the correlation test results clearly show that the age is correlated with the maximum heart rate achieved. The negative value of $r = -0.40227$ indicates that they have a negative linear relationship, meaning that as the age increases, the maximum heart rate decreases. This implies that as people get older, the heart function weakens, leading to a lower heart rate. This decline in heart function with age can be attributed to factors such as reduced elasticity of blood vessels, decreased cardiac output, and other age-related changes in the cardiovascular system.

3.3 Regression Test

We use a regression test to explain the impact of changes in the resting blood pressure (independent variable, x) on the maximum heart rate (dependent variable, y). The sample regression line provides an estimate of the population regression line. The estimated regression equation (estimated regression model), the formula used for calculating the test statistic value, t and degrees of freedom, df are displayed below:

Estimated Regression Model, $\hat{y}_i = b_0 + b_1x$

where,

$\hat{y}_i$ = Estimated dependent variable value

b_0 = Estimate of the regression intercept

b_1 = Estimate of the regression slope

x = Independent variable

Test statistic value, $t = \frac{b_1 - \beta_1}{s_{b_1}}$

where,

b_1 = Sample regression slope coefficient

β_1 = Hypothesized slope

s_{b_1} = Estimator of the standard error of the slope

Degree of freedom, $df = n - 2$

where, n = Sample size

Hypothesis statement:

$H_0 : \beta_1 = 0$ (No linear relationship)

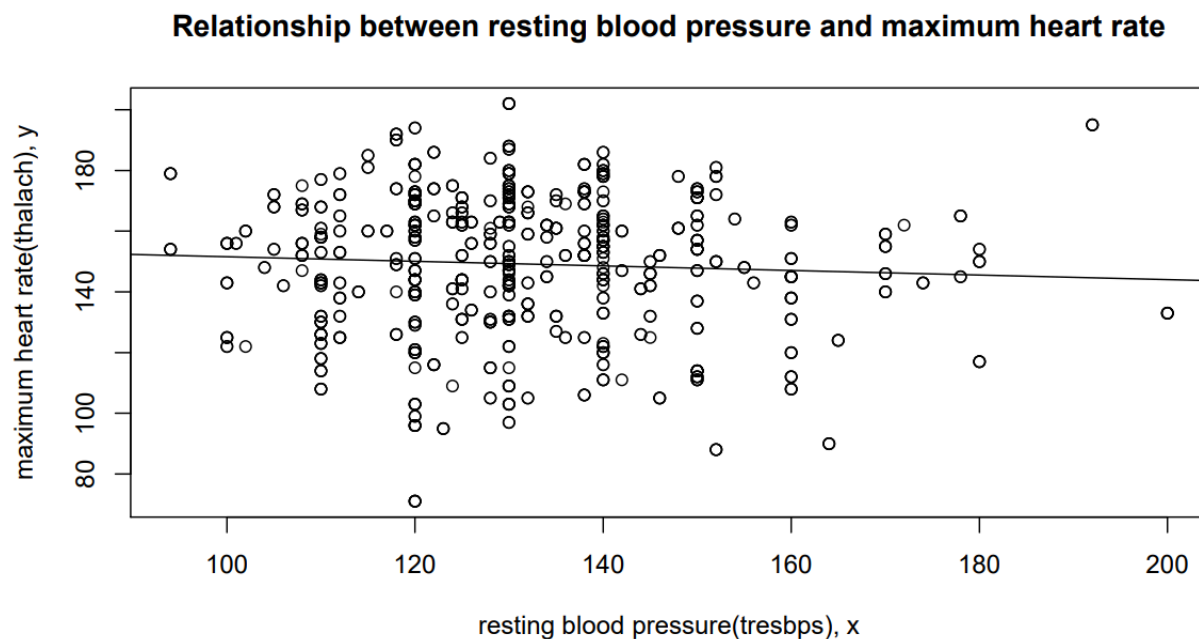
$H_1 : \beta_1 \neq 0$ (Linear relationship exists)

where β_1 represent the hypothesized slope

Test statistic:

A significant level of 0.05 ($\alpha = 0.05$) is used to test the claim that there is a linear relationship between resting blood pressure of the patient and the maximum heart rate achieved by the patient. By using R studio, we have found that the sample regression slope coefficient, $b_1 = -0.07556$, estimate of the standard error of the slope, $s_{b_1} = 0.04567$. By using the formula above and R studio, we also get the test statistic value, $t = -1.655$ and the degree of freedom, $df = 798$,

The critical value that we found is $-t_{\alpha/2, n-2} = -1.962941$ and $t_{\alpha/2, n-2} = 1.962941$ (as it is a two-tailed test, there would be two critical values for this test).



Graph 3: Scatter plot between resting blood pressure and maximum heart rate achieved

Interpretation result:

We have selected a significant level of 0.05 ($\alpha = 0.05$) because we have a large sample size. Since the test statistic value, t has not fallen in the critical region, it is within the range where $t > -t_{\alpha/2, n-2}$ and $t < t_{\alpha/2, n-2}$ thus fail to reject null hypothesis, H_0 . There is 95% strength of evidence to support the claim that there does not exist a linear relationship between resting blood pressure of the patient and maximum heart rate achieved by the patient.

Conclusion and Discussion:

Based on the scatter plot in Graph 3, we can know that it is a negative linear correlation between resting blood pressure of the patient and maximum heart rate achieved by the patient. This means that greater maximum heart rate achieved by the patient, lowers the patient's resting blood pressure. This can also be proven by the sample regression slope coefficient, b_1 which has a negative value.

By using R studio, we can get the value for $b_0 = 159.14622$ (the estimated average value of y when the value of x is 0) and $b_1 = -0.07556$ (the estimated change in the average value of y as a result of one-unit change in x). By substituting the value of both variables in estimated regression model, we can get the estimated regression equation as followed:

$$\hat{y}_i = 159.14622 - 0.07556x$$

b_0 indicates that for individuals within the range of resting blood pressure we observed, 159.14622 is the portion of the maximum heart rate achieved which is not explained by resting blood pressure. Besides, b_1 indicates that the average maximum heart rate decreased by 0.07556 for each additional 1mmHg resting blood pressure.

The coefficient of determination, R^2 is the portion of the variation in the dependent variable that is explained by variation in the independent variable.

$$R^2 = \frac{SSR}{SST}$$

$$= \frac{\text{sum of squares explained by regression}}{\text{total sum of squares}}$$

By using R studio, we can get the value for $R^2 = 0.003419$. Since $0 < R^2 < 1$, we can say that it is a very weak linear relationship between resting blood pressure of the patient and maximum heart rate achieved by the patient. This low R^2 value indicates that resting blood pressure is not a good predictor of maximum heart rate in this dataset.

3.4 Test of Independence

Test of Independence: Chi-square Test for Sex and Chest Pain Type

Chi-Squared Statistic:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Degree of Freedom: $(r - 1) \times (c - 1)$

$$\text{Expected frequency} = \frac{\text{Row total} \times \text{Column total}}{\text{Grand total}}$$

The null hypothesis (H_0) is that sex and chest pain type are independent.

```
> print(paste("Chi-Square Statistic:", chi_square_statistic))
[1] "Chi-Square Statistic: 15.096738778619"
> print(paste("P-value:", p_value))
[1] "P-value: 0.00173581758242018"
> print(paste("Degrees of Freedom:", degrees_of_freedom))
[1] "Degrees of Freedom: 3"
> print("Expected Frequencies:")
[1] "Expected Frequencies:"
> print(expected_frequencies)
```

	0	1	2	3
0	117.475	44.45	74.6125	17.4625
1	252.525	95.55	160.3875	37.5375

The alternative hypothesis (H_1) is that sex and chest pain type are not independent.

	Chest pain type 0	Chest pain type 1	Chest pain type 2	Chest pain type 3
Sex 0	117.475	44.75	74.6125	17.4625
Sex 1	252.525	95.55	160.3875	37.5375

The p-value is 0.00174, which is less than 0.05, suggesting that sex and chest pain type are not independent. Since the p-value (0.00174) is less than the significance level (0.05), we reject the null hypothesis. Therefore, we conclude that there is a significant association between sex and chest pain type. In other words, the type of chest pain experienced is not independent of the sex of the individual.

4.0 Conclusion

In short, there are various factors that influence health outcomes, such as age and sex. Based on our findings, we observed that the mean maximum heart rate decreases with age, indicating a significant negative correlation between age and cardiac performance. Additionally, the Chi-square test revealed a significant association between sex and chest pain type, suggesting that chest pain presentations differ between males and females. Furthermore, we discovered that while there is a noticeable relationship between age and heart rate, the overall influence of age on maximum heart rate is relatively weak. This implies that other factors may also play important roles in determining cardiac health.

It presents an analytical study examining the relationship between various health indicators and demographic factors, using statistical methods to derive conclusions. The correlation and regression analysis revealed a negative linear correlation between age and maximum heart rate, indicating that the maximum heart rate decreases by an average of 1.00554 units for each additional year of age. The coefficient of determination (R^2) was 0.1618, suggesting a weak linear relationship where 16.18% of the variation in maximum heart rate is explained by age. A Chi-square test of independence was conducted to assess the relationship between sex and chest pain type. The p-value of 0.00174, which is less than the significance level of 0.05, led to the rejection of the null hypothesis, indicating a significant association between sex and chest pain type. The study effectively utilized RStudio for statistical analysis and data visualization, demonstrating its utility in computing regression coefficients, constructing regression equations, determining the coefficient of determination, and performing Chi-square tests. The findings underscore the demographic influences on health metrics and offer valuable insights for further research and clinical practices.

5.0 Appendix

Key: Age = age, Sex = sex, Chest pain type = cp, Resting Blood Pressure = trestbps, Maximum Heart Rate = thalach

age	sex	cp	trestbps	thalach
52	1	0	125	168
53	1	0	140	155
70	1	0	145	125
61	1	0	148	161
62	0	0	138	106
58	0	0	100	122
58	1	0	114	140
55	1	0	160	145
46	1	0	120	144
54	1	0	122	116
71	0	0	112	125
43	0	0	132	136
34	0	1	118	192
51	1	0	140	122
52	1	0	128	156
34	0	1	118	192
51	0	2	140	142
54	1	0	124	109
50	0	1	120	162
58	1	2	140	165
60	1	2	140	155
67	0	0	106	142
45	1	0	104	148
63	0	2	135	172
42	0	2	120	173
61	0	0	145	146
44	1	2	130	179
58	0	1	136	152
56	1	2	130	142
55	0	0	180	117
44	1	0	120	144
50	0	1	120	162
57	1	0	130	115
70	1	2	160	112
50	1	2	129	163
46	1	2	150	147
51	1	3	125	125
59	1	0	138	182
64	1	0	128	105
57	1	2	128	150
65	0	2	160	151
54	1	2	120	147
61	0	0	130	169
46	1	0	120	144
55	0	1	132	166
42	1	0	140	178

41	1	1	135	132
66	0	0	178	165
66	0	2	146	152
60	1	0	117	160
58	0	3	150	162
57	0	0	140	123
38	1	2	138	173
49	1	2	120	139
55	1	0	140	111
55	1	0	140	111
56	1	3	120	162
48	1	1	130	180
67	1	2	152	150
57	1	1	154	164
29	1	1	130	202
66	0	2	146	152
67	1	0	100	125
59	1	2	150	157
29	1	1	130	202
59	1	3	170	159
53	1	2	130	152
42	1	0	136	125
37	0	2	120	170
62	0	0	160	145
59	1	0	170	140
61	1	0	140	138
56	1	0	125	144
59	1	0	140	162
48	1	0	130	150
47	1	2	138	156
48	1	2	124	175
63	1	0	140	144
52	1	1	134	158
52	1	1	134	158
50	1	2	140	163
49	1	2	118	126
46	1	2	150	147
38	1	2	138	173
37	0	2	120	170
44	1	1	120	170
58	1	2	140	165
59	0	0	174	143
62	0	0	140	160
68	1	0	144	141
54	0	2	108	167
62	0	0	124	163
63	1	0	140	144

44	1	0	120	144
62	1	1	128	140
45	0	0	138	152
57	0	0	128	159
53	1	0	123	95
65	1	0	110	158
76	0	2	140	116
43	0	2	122	165
57	1	2	150	173
54	1	1	108	156
47	1	2	138	156
52	1	3	118	190
47	1	0	110	118
51	1	0	140	173
62	1	1	120	103
40	1	0	152	181
54	1	0	110	108
44	1	0	110	177
53	1	0	142	111
48	1	0	130	150
57	1	0	110	143
59	1	2	126	134
61	0	0	145	146
63	1	0	130	147
43	1	0	120	120
29	1	1	130	202
42	1	1	120	162
54	1	1	108	156
44	1	0	120	144
60	1	0	145	142
65	0	2	140	157
61	1	0	120	140
60	0	3	150	171
66	1	0	120	151
53	1	2	130	152
52	1	2	138	169
57	1	0	140	148
60	0	3	150	171
51	0	2	130	149
41	1	1	135	132
50	1	2	129	163
54	1	1	108	156
58	0	0	170	146
55	0	1	132	166
64	0	0	180	154
47	1	2	138	156
41	1	1	110	153

57	1	0	152	88
63	0	0	124	136
61	1	3	134	145
34	1	3	118	174
47	1	0	112	143
40	1	0	110	114
51	0	2	120	157
41	1	0	110	158
52	1	3	152	178
39	1	2	140	182
58	1	0	114	140
54	1	1	192	195
58	1	0	125	171
54	1	2	120	147
63	1	0	130	132
54	1	1	108	156
40	1	3	140	178
54	1	2	120	147
67	0	2	115	160
41	1	1	120	182
77	1	0	125	162
51	1	2	100	143
77	1	0	125	162
48	1	0	124	166
56	1	0	125	144
59	1	0	170	140
56	1	0	132	105
57	0	0	120	163
43	1	2	130	162
45	0	1	112	138
43	1	0	150	171
56	1	0	130	103
56	1	1	120	169
39	0	2	94	179
54	1	0	110	126
56	0	0	200	133
56	1	0	130	103
64	1	0	120	96
44	1	0	110	177
56	0	0	134	150
63	1	0	140	144
64	1	3	110	144
60	1	0	140	170
42	1	2	130	150
45	1	1	128	170
57	1	0	165	124
40	1	0	110	114

56	1	0	125	144
63	1	0	130	147
64	1	2	125	131
41	1	2	112	179
56	1	1	130	163
67	0	2	115	160
69	1	3	160	131
67	1	0	160	108
59	1	2	150	157
58	1	0	100	156
45	1	0	115	185
60	0	2	102	160
50	1	0	144	126
62	0	0	124	163
34	1	3	118	174
52	1	3	152	178
64	1	3	170	155
66	0	2	146	152
42	1	3	148	178
59	1	2	126	134
41	1	2	112	179
38	1	2	138	173
62	1	1	120	103
42	1	2	120	194
67	1	0	100	125
50	1	0	150	128
43	1	2	130	162
45	1	1	128	170
49	1	1	130	171
65	1	0	135	127
41	1	1	120	182
46	1	0	140	120
54	1	0	122	116
57	0	1	130	174
63	1	0	130	147
64	1	3	110	144
39	0	2	94	179
51	1	0	140	186
54	1	2	150	165
49	1	2	118	126
44	0	2	118	149
52	1	1	128	184
66	0	0	178	165
58	1	0	125	171
56	1	1	120	178
60	1	0	125	141
41	0	1	126	163

49	0	0	130	163
64	1	3	170	155
49	1	2	118	126
57	1	1	124	141
60	1	0	117	160
62	0	0	150	154
54	0	1	132	159
67	1	2	152	150
38	1	2	138	173
60	1	2	140	155
51	1	2	125	166
44	1	1	130	188
54	1	1	192	195
46	1	0	140	120
39	0	2	138	152
42	1	2	130	150
47	1	0	110	118
45	0	1	112	138
55	1	0	132	132
57	1	0	165	124
35	1	0	120	130
62	0	0	140	157
35	0	0	138	182
64	0	0	180	154
38	1	3	120	182
66	1	0	120	151
44	1	2	120	169
54	1	2	150	165
48	1	0	122	186
55	0	1	132	166
58	0	0	170	146
45	1	0	104	148
53	1	0	123	95
67	1	0	120	71
58	1	2	132	173
71	0	2	110	130
43	1	0	110	161
44	1	1	120	173
39	0	2	138	152
54	1	0	110	108
66	1	0	160	138
56	1	0	130	103
57	1	0	132	168
44	1	1	130	188
55	1	0	160	145
41	0	1	105	168
45	0	1	130	175

35	1	1	122	174
41	0	1	130	172
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71	0	2	110	130
64	0	2	140	133
71	0	1	160	162
58	0	2	120	172
40	1	0	152	181
52	1	2	138	169
58	1	0	128	130
61	1	2	150	137
59	1	2	150	157
56	0	0	200	133
67	1	0	100	125
67	1	0	120	71
58	1	0	150	111
35	1	1	122	174
52	1	1	120	172
46	0	1	105	172
51	1	2	94	154
55	0	1	132	166
60	1	0	145	142
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62	1	0	120	99
44	0	2	118	149
44	1	1	120	170
59	1	2	126	134
56	0	1	140	153
61	1	0	120	140
48	1	0	130	150
70	1	2	160	112
74	0	1	120	121
40	1	3	140	178
42	1	3	148	178
64	0	2	140	133
63	0	2	135	172
59	1	0	140	162
53	0	2	128	115
53	0	0	130	143
48	0	2	130	139
45	1	0	142	147
66	1	1	160	120
48	1	1	130	180
56	0	1	140	153
54	1	1	192	195
57	1	0	150	112

70	1	0	130	109
53	0	2	128	115
37	0	2	120	170
63	0	0	108	169
37	1	2	130	187
54	0	2	110	158
60	1	0	130	132
58	1	0	150	111
57	1	2	150	173
54	1	2	125	152
56	1	2	130	142
60	1	0	130	144
38	1	2	138	173
44	1	2	120	169
65	0	2	155	148
52	1	2	172	162
41	1	1	120	182
66	1	1	160	120
50	1	0	150	128
54	0	2	108	167
43	1	0	132	143
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66	1	0	120	151
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57	1	0	110	143
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57	1	1	124	141
46	0	0	138	152
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67	1	0	160	108
59	1	3	134	162
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62	1	2	130	146
43	0	2	122	165
53	1	2	130	173
57	0	1	130	174
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58	1	2	112	165
48	1	1	110	168
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51	1	2	110	123
43	0	0	132	136
55	1	0	132	132
54	0	2	110	158
58	1	1	120	160
46	0	2	142	160

66	1	0	160	138
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64	0	0	130	122
67	1	0	120	71
52	1	3	118	190
58	1	0	146	105
58	1	2	132	173
59	1	0	110	142
58	1	0	150	111
35	1	0	126	156
51	1	2	110	123
42	0	2	120	173
77	1	0	125	162
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58	0	1	136	152
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51	1	2	110	123
62	0	0	160	145
63	1	0	130	132
66	0	2	146	152
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66	1	0	160	138
63	1	3	145	150
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71	0	0	112	125
70	1	1	156	143
46	0	1	105	172
61	1	0	140	138
56	1	2	130	142
58	1	2	140	165
58	1	0	100	156
46	0	0	138	152
46	1	2	150	147
41	0	1	105	168
56	1	0	125	144
57	1	0	150	112
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65	0	2	160	151
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51	0	0	130	142
54	0	2	160	163
57	1	2	150	174
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47	1	2	108	152
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65	0	0	150	114
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41	0	2	112	172
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47	1	2	130	179
58	1	1	120	160
62	0	0	150	154
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57	1	0	152	88
57	1	2	150	174
47	1	2	130	179
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55	1	0	160	145
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63	0	0	150	154
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69	1	2	140	146
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47	1	0	110	118
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48	1	1	110	168
57	1	2	150	173
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62	0	0	124	163
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54	1	2	125	152
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70	1	0	130	109
67	0	0	106	142
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54	1	0	122	116
43	0	0	132	136
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67	1	0	125	163
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39	0	2	94	179
67	1	0	120	71
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42	0	2	120	173

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43	1	0	120	120
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41	1	1	110	153
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60	1	0	130	144
63	1	0	130	147
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68	1	2	180	150
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60	1	2	140	155
63	0	0	150	154
52	1	0	128	156
58	1	0	125	171
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55	0	1	135	161
53	1	0	140	155
63	0	1	140	179
39	1	0	118	140
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50	0	2	120	158
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64	1	0	120	96
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45	0	0	138	152
62	0	0	150	154
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64	0	0	130	122
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54	1	0	110	126
54	1	0	110	126
68	1	0	144	141
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62	1	1	128	140
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Table 3: Raw Dataset of The Analysis